Background Exercises

AE 513: Multivariable Control — Autumn 2025

Linear Algebra

Q1. Let $A = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$. What are the eigenvalues of A?

(a) $\{4, 2\}$
(b) $\{6, -1\}$
(c) $\{1, 1\}$
(d) $\{4, 1\}$

Q2. Let $B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Which vector is an eigenvector with eigenvalue 4?

(a) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
(b) $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
(c) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
(d) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Q3. Let $A \in \mathbb{R}^{2\times 3}$, $B \in \mathbb{R}^{3\times 2}$, $x \in \mathbb{R}^2$. Which statement is correct?

(a) $AB \in \mathbb{R}^{3\times 3}$; $BA \in \mathbb{R}^{2\times 2}$; Ax defined; Bx undefined.
(b) $AB \in \mathbb{R}^{2\times 2}$; $BA \in \mathbb{R}^{3\times 3}$; Ax *undefined*; $Bx \in \mathbb{R}^3$.
(c) $AB \in \mathbb{R}^{2\times 3}$; $BA \in \mathbb{R}^{3\times 2}$; both Ax, Bx defined.
(d) Only Ax is defined.

Q4. Which identity holds for any conformable A, B?

(a) $(AB)^\top = A^\top B^\top$
(b) $(AB)^\top = B^\top A^\top$
(c) $(AB)^\top = AB$
(d) None of the above

Q5. Determine the definiteness of the matrix $D = \mathrm{diag}(2, 5)$.

(a) Positive-Semi-Definite
(b) Positive-Definite
(c) Negative-Definite
(d) Indefinite

Q6. For any real M, which statements are always true?

(a) $M + M^\top$ is symmetric
(b) $M - M^\top$ is symmetric
(c) $(M + M^\top)^\top = M + M^\top$
(d) Both (a) and (c)

Q7. Let T be invertible and $C = TAT^{-1}$. Which are always equal for A and C?

(a) trace and determinant
(b) trace only
(c) determinant only
(d) all entries

Q8. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. What is $\mathrm{rank}(A)$?

(a) 0
(b) 1
(c) 2
(d) 3

Q9. Which matrix is orthogonal?

(a) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
(b) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
(d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Q10. What is D^{-1} for $D = \mathrm{diag}(2, 5)$?

(a) $\mathrm{diag}(2, 5)$
(b) $\mathrm{diag}(5, 2)$
(c) $\mathrm{diag}\left(\frac{1}{5}, \frac{1}{2}\right)$
(d) $\mathrm{diag}\left(\frac{1}{2}, \frac{1}{5}\right)$

Multivariable Calculus

Q11. For $f(x, y) = x^2 y$, compute $\nabla f(x, y)$.

(a) $(x^2,\ 2xy)$
(b) $(2xy,\ x^2)$
(c) $(2x,\ y)$
(d) $(2x,\ 2y)$

Q12. For $f(x, y) = e^x y^2$, compute $\nabla f(x, y)$.

(a) $(y^2 e^x,\ 2ye^x)$
(b) $(e^x,\ 2y)$
(c) $(y^2,\ 2y)$
(d) $(y^2,\ 2ye^x)$

Q13. For $F(x, y) = \left(x^2,\ xy\right)$, find the Jacobian $J_F(x, y)$.

(a) $\begin{bmatrix} 2x & 0 \\ y & x \end{bmatrix}$

(b) $\begin{bmatrix} 2x & y \\ 0 & x \end{bmatrix}$

(c) $\begin{bmatrix} x & 0 \\ y & 2x \end{bmatrix}$

(d) $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

Q14. For $G(x, y, z) = \left(x + y,\ x + z\right)$, compute $J_G(x, y, z)$.

(a) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Q15. For $f(x,y) = 3x^2 + 2xy + y^2$, compute the Hessian $Hf(x,y)$.

(a) $\begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & 1 \\ 2 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 6 & 2 \\ 1 & 2 \end{bmatrix}$

Q16. For $f(x,y) = x^3 + y^2$, compute the Hessian $Hf(x,y)$.

(a) $\begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 3x^2 & 0 \\ 0 & 2y \end{bmatrix}$ (d) $\begin{bmatrix} 6x & 3x^2 \\ 0 & 2 \end{bmatrix}$

Q17. For $f(x,y) = x^2 + y^2$, find the critical point(s) and classify.

(a) No critical points; function increases everywhere
(b) Single critical point at $(0,0)$; strict local minimum
(c) Single critical point at $(0,0)$; strict local maximum
(d) Single critical point at $(0,0)$; saddle point

Q18. For $f(x,y) = xe^y$, compute $\dfrac{\partial f}{\partial x}(1,0)$.

(a) 0
(b) 1
(c) e
(d) e^1

Q19. Let $x = r\cos\theta,\ y = r\sin\theta$. For $f(x,y) = x^2 + y^2$, compute $\dfrac{\partial}{\partial r} f(x(r,\theta), y(r,\theta))$.

(a) 0
(b) r
(c) 2
(d) $2r$

Q20. For the map $(u,v) \mapsto (x,y) = (2u,\ 3v)$, compute $\det\big(\partial(x,y)/\partial(u,v)\big)$.

(a) 0
(b) 1
(c) 6
(d) -6

Optimization

Q21. Minimize $f(x) = x^2 + 2x + 1$ over $x \in \mathbb{R}$. What is the minimizer x^*?

(a) -1
(b) 0
(c) 1
(d) No minimizer

Q22. Minimize $f(x, y) = (x-1)^2 + (y+2)^2$ over $\mathbb{R}^2$. What is (x^*, y^*)?

(a) $(1, -2)$
(b) $(-1, 2)$
(c) $(1, 2)$
(d) $(0, -2)$

Q23. For $f(x, y) = x^2 + y^2 + x$, solve $\nabla f = 0$ and give the minimizer.

(a) $\left(\frac{1}{2}, 0\right)$
(b) $\left(-\frac{1}{2}, 0\right)$
(c) $(0, 0)$
(d) $(0, -\frac{1}{2})$

Q24. For $f(x) = ax^2 + bx + c$ with $a > 0$, which statement is true?

(a) f is concave on $\mathbb{R}$ and has no minimizer
(b) f is convex on $\mathbb{R}$ and has a unique global minimizer $x^* = -\dfrac{a}{2b}$
(c) f is convex on $\mathbb{R}$ and has a unique global minimizer $x^* = -\dfrac{b}{2a}$
(d) f is neither convex nor concave

Q25. Minimize $(x-3)^2$ subject to $x \geq 5$. What is the minimizer x^*?

(a) 3
(b) 4
(c) 5
(d) No minimizer

Q26. Which function is convex on $\mathbb{R}$?

(a) $-x^2$
(b) e^x
(c) x^3
(d) $\sin x$

Basic Python

Q27. What does this print?

```
def add(a, b):
    print(a + b)

x = add(2, 3)
print(x)
```

(a) Prints `5` only
(b) Prints `None` only
(c) Prints `5` then `None`
(d) Raises an error

Q28. Let `nums = [10, 20, 30, 40, 50]`. What is `nums[4:0:-2]`?

(a) `[50, 30]`
(b) `[50, 30, 10]`
(c) `[40, 20]`
(d) Error

Q29. What does this print?

```
def append_item(x, lst=[]):
    lst.append(x)
    return lst

print(append_item(1))
print(append_item(2))
```

(a) `[1]` then `[2]`
(b) `[1]` then `[1, 2]`
(c) Two separate lists: `[1]` then `[2]`
(d) Error

Q30. What does this print?

```
m = [[0]*3]*2
m[0][1] = 9
print(m)
```

(a) `[[0, 9, 0], [0, 0, 0]]`
(b) `[[0, 9, 0], [0, 9, 0]]`
(c) `[[0, 0, 0], [0, 0, 0]]`
(d) Error

Q31. What does this print?

```
a = [1, 2, 3]
b = a
a.append(4)
print(b)
```

(a) `[1, 2, 3]`
(b) `[1, 2, 3, 4]`
(c) `None`
(d) Error

Q32. (Column vector) You have `v = [1, 2, 3]`. Create a NumPy column vector with shape `(3, 1)`. Which snippet does that?

(a) `np.array(v)`
(b) `np.reshape(v, (1, 3))`
(c) `np.array(v).T`
(d) `np.array(v).reshape(-1, 1)`

Q33. (Matmul shape) If `A.shape == (2, 3)` and `x.shape == (3,)`, what is the shape of `A @ x`?

(a) `(2,)`
(b) `(2, 1)`
(c) `(1, 2)`
(d) Error